COMPUTER NETWORK

UNIT-WISE IMPORTANT QUESTIONS

Based on analysis of the three provided previous-year papers and the supplied syllabus

★ = Highest-priority preparation question

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UNIT 1: Introduction to Computer Networks

1. ★ Explain the OSI Reference Model with a neat diagram and describe the functions of all seven layers.
2. ★ Explain the TCP/IP Reference Model with a neat diagram and describe the functions of its layers.
3. ★ Differentiate between the OSI Model and the TCP/IP Model.
4. Explain the concept of computer networks, their advantages, and applications.
5. Explain how data travels through the different layers of the OSI Model.
6. Explain the working of a network simulation tool such as Wireshark or Cisco Packet Tracer.

UNIT 2: Data Communication and Communication Methodologies

1. ★ Explain different types of network topologies with neat diagrams, advantages, and disadvantages.
2. ★ Differentiate between guided and unguided transmission media with suitable examples.
3. ★ Explain different switching techniques: Circuit Switching, Packet Switching, and Message Switching.
4. Explain different types of transmission media: Coaxial Cable, UTP, STP, Optical Fibre, Microwave, and Satellite.
5. Differentiate between UTP and STP cables. Explain their features and applications.
6. Explain the functions of the Data Link Layer and discuss Ethernet, WLAN, and Bluetooth protocols.

UNIT 3: Network Layer

1. ★ Explain IPv4 addressing, its structure, and different IP address classes with examples.
2. ★ Differentiate between IPv4 and IPv6. Explain the important features of IPv6.
3. ★ What is routing? Explain the Distance Vector Routing algorithm with a suitable example.
4. Explain Link-State Routing and compare it with Distance Vector Routing.
5. Explain the working of RIP and OSPF routing protocols and differentiate between them.
6. Explain subnetting and demonstrate how an IP network can be divided into smaller subnets.

UNIT 4: Transport Layer

1. ★ Differentiate between TCP and UDP protocols with their features, advantages, and applications.

2. ★ What is DNS? Explain its role and working in Internet communication.
3. ★ What is SMTP? Explain its working and role in sending emails.
4. Explain the functions and design issues of the Transport Layer.
5. Explain error control and flow control mechanisms in the Transport Layer.
6. Explain TCP communication between two devices with a suitable example.

UNIT 5: Network Devices

1. ★ Explain the functions of Hub, Bridge, Switch, and Router. Differentiate between them.
2. ★ Explain the working, functions, and applications of a Network Interface Card (NIC).
3. ★ What is SNMP? Explain its architecture, working, and applications in network management.
4. Explain the working of Wi-Fi devices and wireless access points.
5. Differentiate between Repeater, Modem, and Gateway, explaining their functions.
6. Explain the role of a Network Management System (NMS) and its importance in managing computer networks.

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